

[INTERNAL]

Qdrant Deployment Strategy — Production Cluster Guide

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Last Updated: 2026-03-22
Classification: internal

This document describes NovaTech's production deployment strategy for Qdrant v1.10 on AWS EKS. It covers cluster sizing, collection design, backup procedures, and the hybrid retrieval configuration powering Project Atlas's Q&A; feature. Following the March 2026 security incident, network policies and mTLS between the API Gateway and Qdrant have been mandated.

1. Cluster Architecture

NovaTech runs a 3-node Qdrant cluster on EKS using r6g.2xlarge instances (64 GB RAM, 8 vCPU). Each node stores a shard replica; all shards are replicated twice for fault tolerance. The cluster serves approximately 2.4 million document chunks ingested from Stack Overflow, Confluence, and internal PDF corpora as of April 2026.

Collection	Vectors	Dim	Distance	Payload Filters
eng_docs	840 000	768	Cosine	dept, classification
prod_docs	620 000	768	Cosine	dept, classification
conf_docs	180 000	768	Cosine	dept, classification
stackoverflow	760 000	768	Cosine	tags, score

2. Hybrid Retrieval Configuration

Each collection stores both a **dense** vector (BAAI/bge-base-en-v1.5) and a **sparse** BM25 vector (computed via FastEmbed's BM25 tokeniser). At query time, Qdrant executes both searches in parallel and the Retriever service merges results using Reciprocal Rank Fusion with k=60. Empirical tests on our internal evaluation set show hybrid retrieval improves NDCG@10 by 18% over dense-only.

- Dense top-k: 40 results per query.
- Sparse top-k: 40 results per query.
- Post-RRF candidates: 80; passed to CrossEncoder reranker, top-5 returned.
- Reranker model: BAAI/bge-reranker-base, served on a g4dn.xlarge GPU node.

3. Backup and Recovery

Qdrant snapshots are taken every 6 hours using the /snapshots REST API and uploaded to s3://novatech-qdrant-backups/. The snapshot Lambda function was introduced after the March 2026 security incident revealed a 14-hour gap in recovery point objective coverage.

Restore SLA: < 2 hours for a full cluster restore from S3 snapshot, validated quarterly via Game Day exercises. Last Game Day: 2026-04-05, restore time: 1h 44m.

4. Security Hardening (Post March 2026 Incident)

Following the unauthorised read access discovered in March 2026, the following controls were implemented within 72 hours: mTLS enforced on all inter-pod communication, Qdrant API key rotated, Kubernetes NetworkPolicy restricting Qdrant port 6333 to the Retriever service namespace only.

- mTLS via cert-manager with 90-day certificate rotation.
- Qdrant API key stored in AWS Secrets Manager; rotated every 30 days.
- Network Policy: only retriever-svc pods in retriever-ns may reach qdrant:6333.
- Audit logging enabled; logs shipped to CloudWatch Logs with 90-day retention.